

Q1

known f is strongly convex, for $x, y \in S$

$$f(y) \geq f(x) + \nabla f(x)^T (y-x) + \frac{m}{2} \|y-x\|_2^2 \quad (1) \text{ (from Lecture 9)}$$

$$\nabla^2 f(x) \geq mI(z) \text{ (from Question)}$$

Prove: $f(x) - p^* \leq \frac{1}{2m} \|\nabla f(x)\|_2^2$

① Let $g(y) = f(x) + \nabla f(x)^T (y-x) + \frac{m}{2} \|y-x\|_2^2$

Let $\nabla g(y) = 0 \Leftrightarrow m\|y-x\| + \nabla f(x)^T = 0 \Leftrightarrow \|y-x\| = -\frac{1}{m} \nabla f(x)^T$
(3)

② Substitute (3) into (1)

$$f(y) \geq f(x) + \nabla f(x)^T \left[-\frac{1}{m} \nabla f(x)^T\right] + \frac{m}{2} \left(-\frac{1}{m}\right)^2 \|\nabla f(x)\|_2^2$$

$$\geq f(x) - \frac{1}{2m} \|\nabla f(x)\|_2^2$$

$\Leftrightarrow f(x) - f(y) \leq \frac{1}{2m} \|\nabla f(x)\|_2^2$

if $p^* = f(y)$, above also establish

Hence, $f(x) - p^* \leq \frac{1}{2m} \|\nabla f(x)\|_2^2$



f is strongly convex on $S = \{x | f(x) \leq f(x^0)\}$

Suppose exact line search is used in gradient descent

Prove:

$$f(x^k) - p^* \leq C^k [f(x^0) - p^*]$$

① According to Taylor's theorem:

$$f(y) = f(x) + \nabla f(x)^T (y-x) + \frac{1}{2} (y-x)^T \nabla^2 f(\xi) (y-x)$$

$$\text{let } y = x^+ = x - t \nabla f(x) \quad [\Delta x = -\nabla f(x)], \quad (x^+ - x) = -t \nabla f(x)$$

$$(1) \quad f(x^+) = f(x) + \nabla f(x)^T (x^+ - x) + \frac{1}{2} (x^+ - x)^T H(\xi) (x^+ - x)$$

$$(2) \quad mI \leq H(\xi) \leq MI \quad (m \text{ is minimum eigenvalue})$$

M is the maximum

substitute (2) into (1)

$$(3) \quad f(x^+) \leq f(x) - t \nabla f(x)^T \nabla f(x) + \frac{Mt^2}{2} \nabla f(x)^T \nabla f(x)$$

$$(2) \quad \text{let } g(t) = -t^2 \nabla f(x)^T \nabla f(x) + \frac{Mt^2}{2} \nabla f(x)^T \nabla f(x) + f(x)$$

$$g'(t) = 0 \Leftrightarrow \hat{t} = 1/M$$

$\Rightarrow$ substitute $\hat{t} = 1/M$ into (3)

$$f(x^+) \leq f(x) - \frac{1}{2M} \nabla f(x)^T \nabla f(x)$$

$$(4) \quad f(x^+) - p^* \leq f(x) - \frac{1}{2M} \nabla f(x)^T \nabla f(x) - p^*$$

Combine (4) with $\nabla f(x)^T \nabla f(x) \geq mI \Leftrightarrow \frac{1}{M} \nabla f(x)^T \nabla f(x) \leq \frac{m}{2M}$

$$(3) \quad \text{from Q1: } \|\nabla f(x)\|^2 \geq 2m [f(x) - p^*]$$

Combine with (4):

$$f(x^+) - p^* + \frac{1}{2M} \nabla f(x)^T \nabla f(x) \leq f(x) - p^* + -\frac{m}{2M} [f(x) - p^*]$$

$$\Rightarrow f(x^+) - p^* \leq (1 - \frac{m}{M}) [f(x^0) - p^*], \quad C = (1 - \frac{m}{M})$$



No.

Q3

Exhaustive line search:

Date.

$$f(x) = \frac{1}{2}(x_1^2 + r x_2^2) \quad \nabla f(x) = (x_1, r x_2)$$

$$x^{k+1} = x^k + \alpha \Delta x, \quad \Delta x = -\nabla f(x^k) \Rightarrow x^{k+1} = x^k - \alpha \nabla f(x^k)$$

$$g(\alpha) = f[x^k - \alpha \nabla f(x^k)] = \frac{1}{2} [(x_1^k - \alpha x_1^k)^2 + r(x_2^k - \alpha r x_2^k)^2]$$

$$\text{Let } g'(\alpha) = -2(1-\alpha)(x_1^k)^2 - 2r^2(1-\alpha r)(x_2^k)^2 = 0$$

$$\Leftrightarrow \alpha = (x_1^k)^2 + r^2(x_2^k)^2 / [(x_1^k)^2 + r^2(x_2^k)^2] \quad (1)$$

known $x_1^0 = r, x_2^0 = 1$, substitute into (1)

$$\alpha = \frac{r^2 + r^2}{r^2 + r^3} = \frac{2}{1+r}, \quad 1-\alpha = \frac{r-1}{r+1}$$

$$\text{Hence, } x_1^k = (1-\alpha)x_1^{k-1} = \left(\frac{r-1}{r+1}\right)^k x_0 = r \left(\frac{r-1}{r+1}\right)^k$$

$$x_2^k = (1-\alpha r)x_2^{k-1} = \left(\frac{r-1}{r+1}\right)^k, \quad \text{Proved}$$

Newton's descent:

$$x^{k+1} = x^k - H^{-1} \nabla f(x^k), \quad \nabla f(x^k) = [x_1^k, r x_2^k]$$

$$H = \nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & r \end{bmatrix}$$

$$H^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/r \end{bmatrix}$$

$$\Rightarrow x^{k+1} = \begin{bmatrix} x_1^k \\ x_2^k \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1/r \end{bmatrix} \begin{bmatrix} x_1^k \\ r x_2^k \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

only one iteration makes $x^0 = (r, 1) \Rightarrow x^1(0, 0)$.



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